AIC and BIC

AIC = **Akaike Information Criterion** = is a statistical matrix used to compare and select the best performance model from a set of candidate models.
It evaluates model quality by balancing goodness-of-fit against model complexity, penalizing models with more parameters to prevent overfitting. A lower AIC value indicates a better-fitting model.

Key Aspects of AIC

- **Formula:** $AIC = 2K - 2\ln(L)$, where K is the number of independent variables (parameters) and L is the maximum likelihood estimate of the model.
- **Model Selection:** When comparing models, the one with the lowest AIC is preferred, as it suggests the best trade-off between accuracy and simplicity.
- **Overfitting Prevention:** By adding a penalty of $2K$ for additional parameters, AIC prevents selecting overly complex models that fit training data well but predict poorly.
- **Information Loss:** AIC is rooted in information theory; it estimates the relative information lost by a given model —the less information lost, the better the model.
- **Application:** It is widely used for model selection, such as in regression analysis, time series analysis, and identifying the best-fit distribution.
- **Limitations:** AIC indicates the *best* model among the set, but not whether the model is good in an absolute sense (e.g., all models could be poor). Wikipedia +6

Note: Cp = AIC
BIC = Bayesian Information Criterion = is a statistical metric for model selection that balances goodness-of-fit with model complexity to prevent overfitting. Larger dataset
The model with the lowest BIC is preferred.

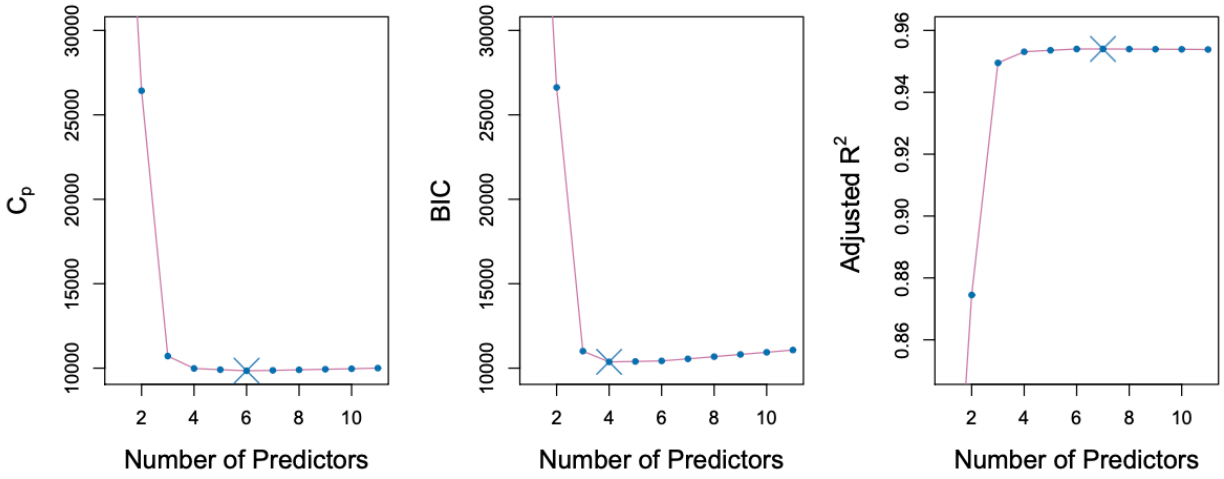
Key Aspects of BIC

- **Formula:** $BIC = k\ln(n) - 2\ln(\hat{L})$.
- **Purpose:** To find the most parsimonious model that fits the data well without unnecessary complexity.
- **Decision Rule:** Lower BIC values indicate a better model.
- **Penalty Term:** The term $k\ln(n)$ acts as a penalty for adding more parameters, which is stricter than AIC for samples $n > 7$.
- **Origin:** Developed by Gideon E. Schwarz in 1978, it serves as an asymptotic approximation to a transformation of the Bayesian posterior probability of a candidate model. Wikipedia +7

AISL
AIC = Cp

$$\mathrm{AIC} = \frac{1}{n}\left(\mathrm{RSS} + 2d\hat{\sigma}^2\right),$$

$$\mathrm{BIC} = \frac{1}{n}\left(\mathrm{RSS} + \log(n) d \hat{\sigma}^2\right).$$



There are two formula for AIC